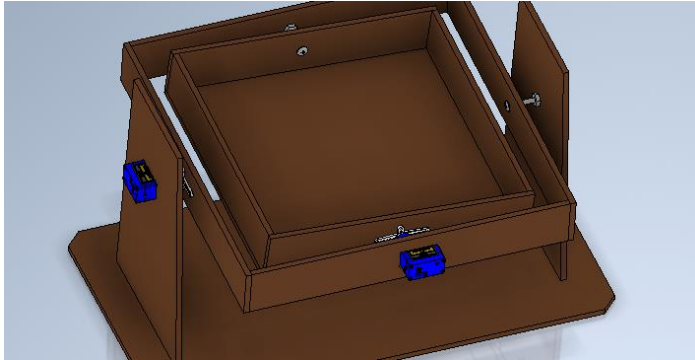


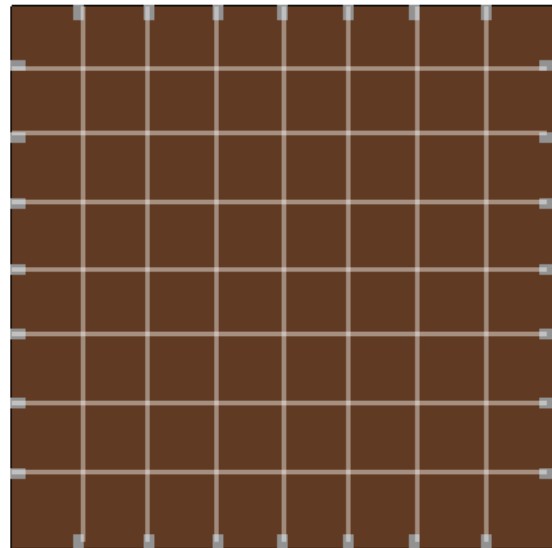
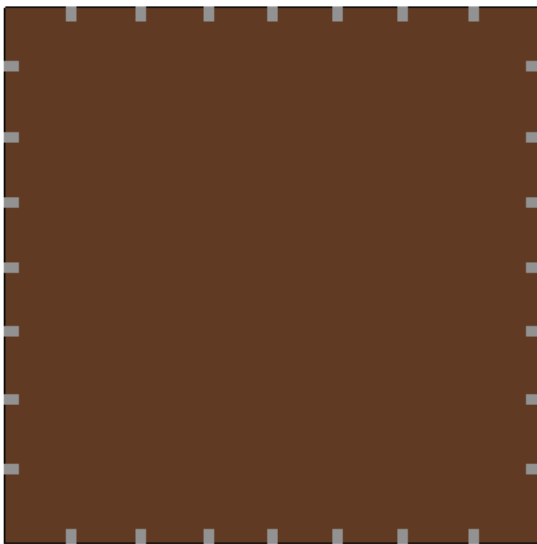
Marble Maze Build Guide



This build guide will lead you through the steps to assemble your marble maze project. Pay close attention to each step as there are some pieces that are *slightly asymmetrical* and if you get their orientation wrong things won't line up correctly!

Step One – Plan out maze pattern

Using the **maze square** piece (16cm x 16cm), mark out a grid every 2cm along each edge. If your school has a laser cutter, you might have these marks already on your base piece.

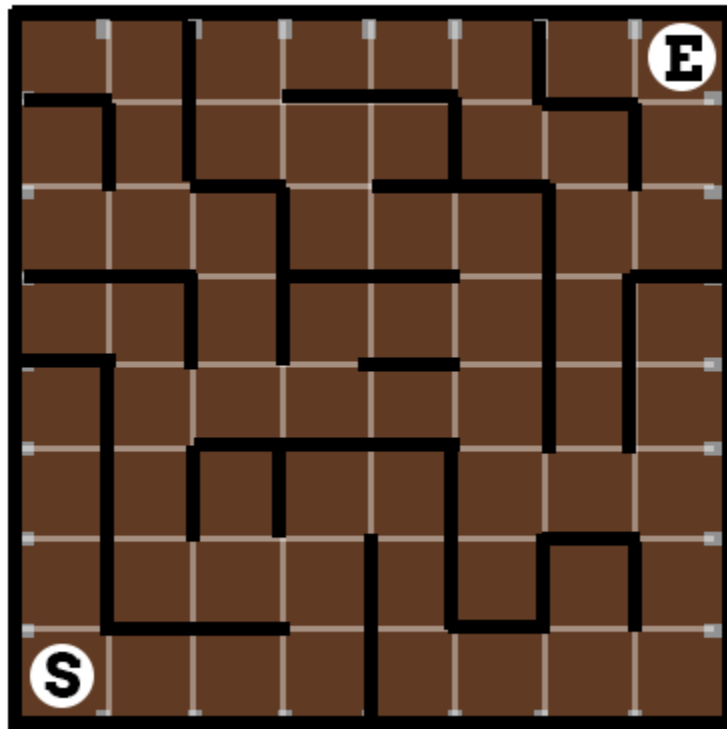


Once you've marked the outside edges, use a **light pencil** to mark out a grid.

Using the grid, you'll plan out where your maze walls will be. A few things to consider:

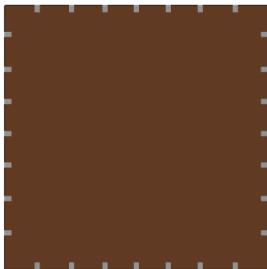
- Maze will be fully contained on the outside walls (so that the marble can't fall out), but you'll specify **starting** and **ending** locations.
- You may want to roughly plan out where the walls will be by darkening your pencil grid with a bit more pressure.

- Once you have your plan, trace the plan in **dark felt marker**.



Step Two – Add Outer Walls

Before we add the *maze walls*, start by building the four outer walls for the maze itself. Pieces you'll need:



16x16cm square base

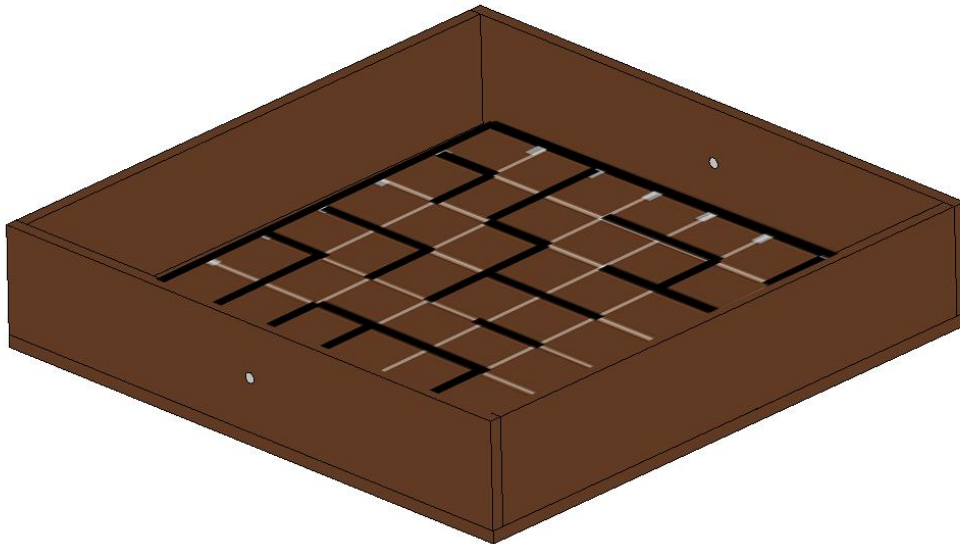


16x3cm outer wall x2



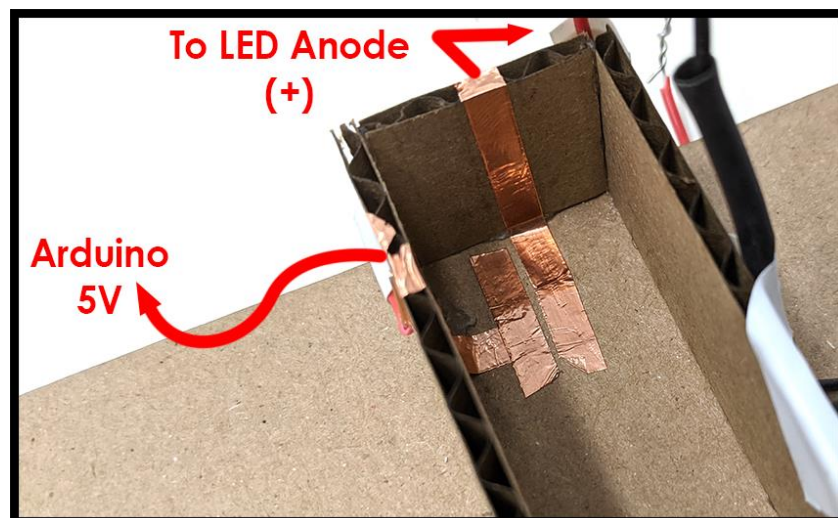
16x3cm outer wall with centerpoint x2

Using hot glue, affix the four walls onto the outside of the square base. Make sure the two walls with centerpoints are **opposite one another**.



Step Three – Prep Expert Challenge

Although we normally would move on to adding walls next, the expert challenge of adding an LED circuit to indicate when the marble reaches the end is a bit tricky to install once the full maze has been built. Instead, you'll do a little prep work here, so that if you are able to attempt that challenge at the end, you'll be set up to do so.

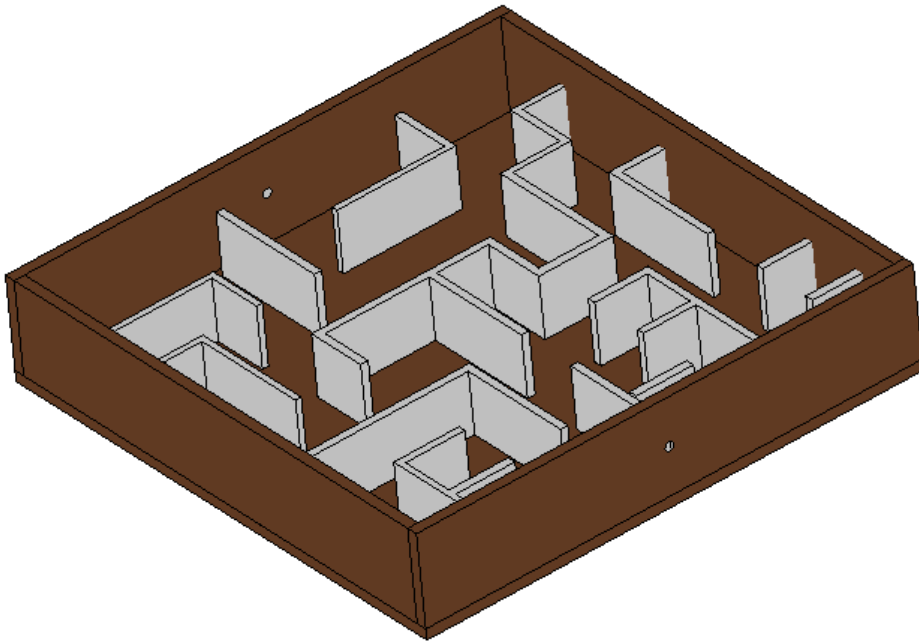


Wherever you've chosen as the **end location**, use a small amount of copper tape to create two parallel paths. They should be close together, but not in contact. For each section, bring the

tape up over the outer wall. This will give us two connections to use later on with the expert challenge.

Step Four – Complete Interior Maze Walls

The next step is likely the longest task to complete. Using 2cm high walls (various lengths will be pre-cut, or you can cut to length yourself), glue pieces down over top of each wall location you marked previously with **dark felt marker**.



Once complete, you should have something similar to this.

Step Five – Build Outer Frame

The next step is to assemble the outer frame. You'll need the following pieces:

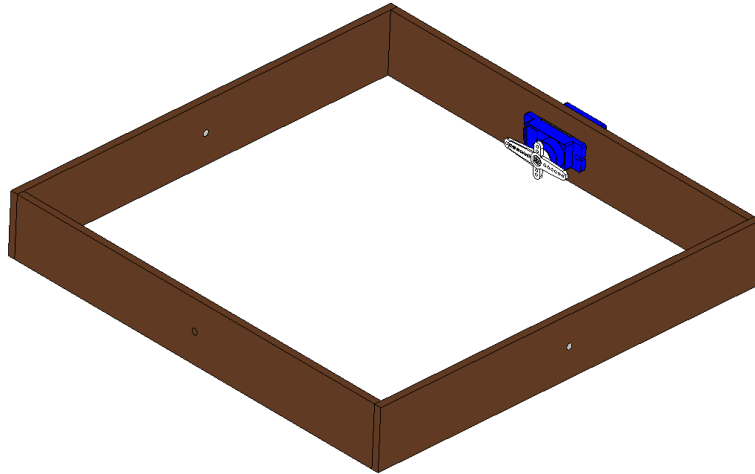


19x3cm frame with centerpoint



19x3cm frame with servo cutout

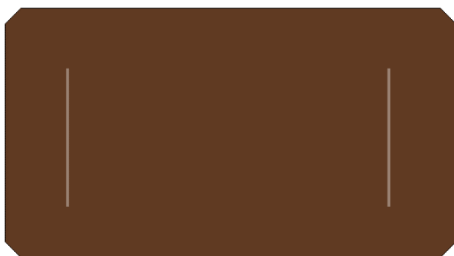
Note: Please notice that the servo cut-out is **not centered**. This is because the output shaft of the servo is offset from the center of its body. When you go to install the servo, please make sure you've oriented the Servo so that the output shaft is centered in the frame.



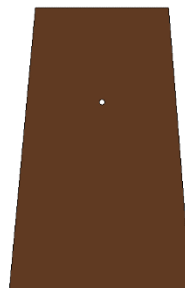
These four pieces can be hot-glued together. Once complete, one of your two servos can be placed into the cut-out as shown above.

Step Six – Prepare Base

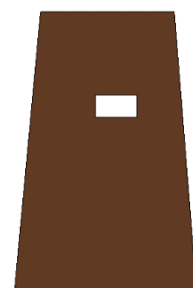
Next we'll make the platform to support the maze pieces. For this step, you'll need the following:



Main Base Plate



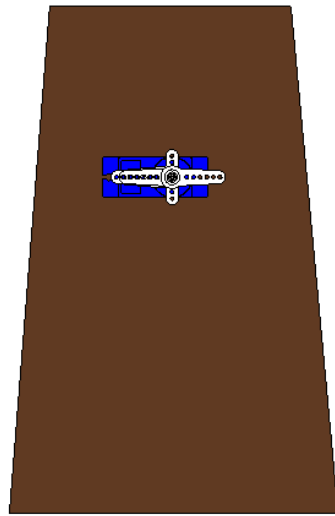
Side Support with Centerpoint



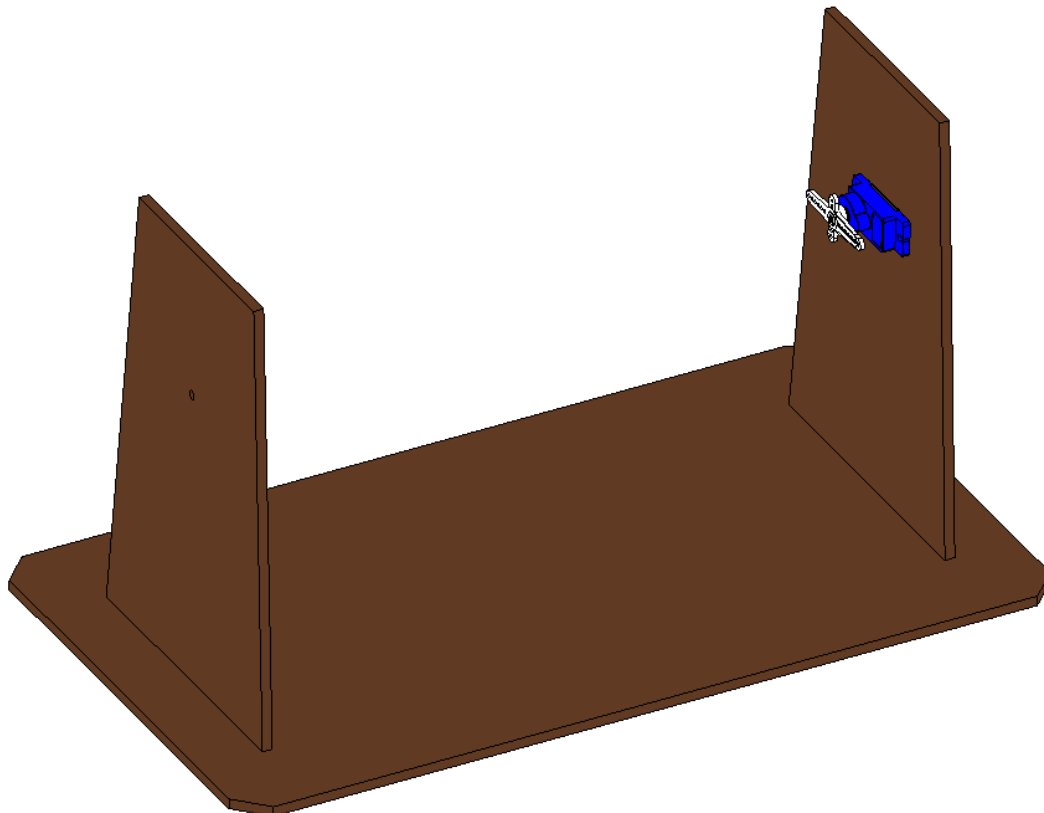
Side Support with Servo cutout

Glue the two side supports onto the base *if you have registration marks specifying how far apart they should be*. (See two lines marked on illustration above)

If you don't have registration marks, skip to assembling the frames and servos together first (next step) so that you can base your support spacing on that piece.



*As with our previous servo, remember that the cutout is not centered, which allows for the output shaft **to be centered**. Be careful that you orient your servo so that this is the case.*



Step Seven – Final Assembly

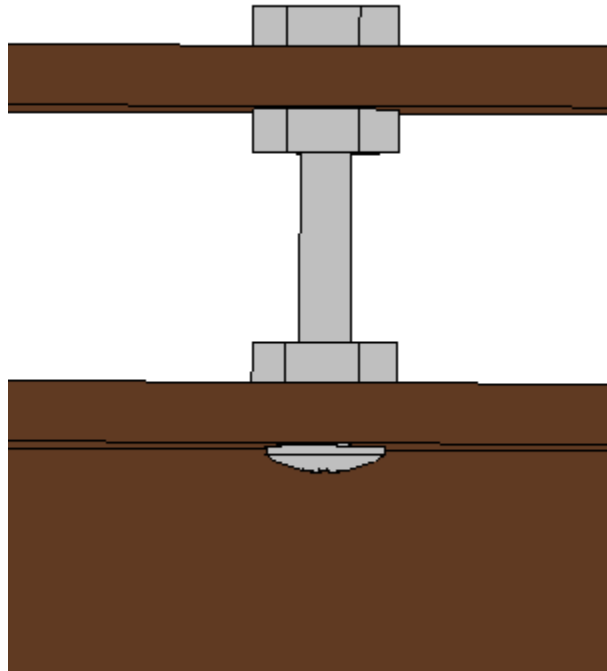
You should now have 3 assemblies:

- **Inner Maze**
- **Outer Frame**
- **Base with Supports**

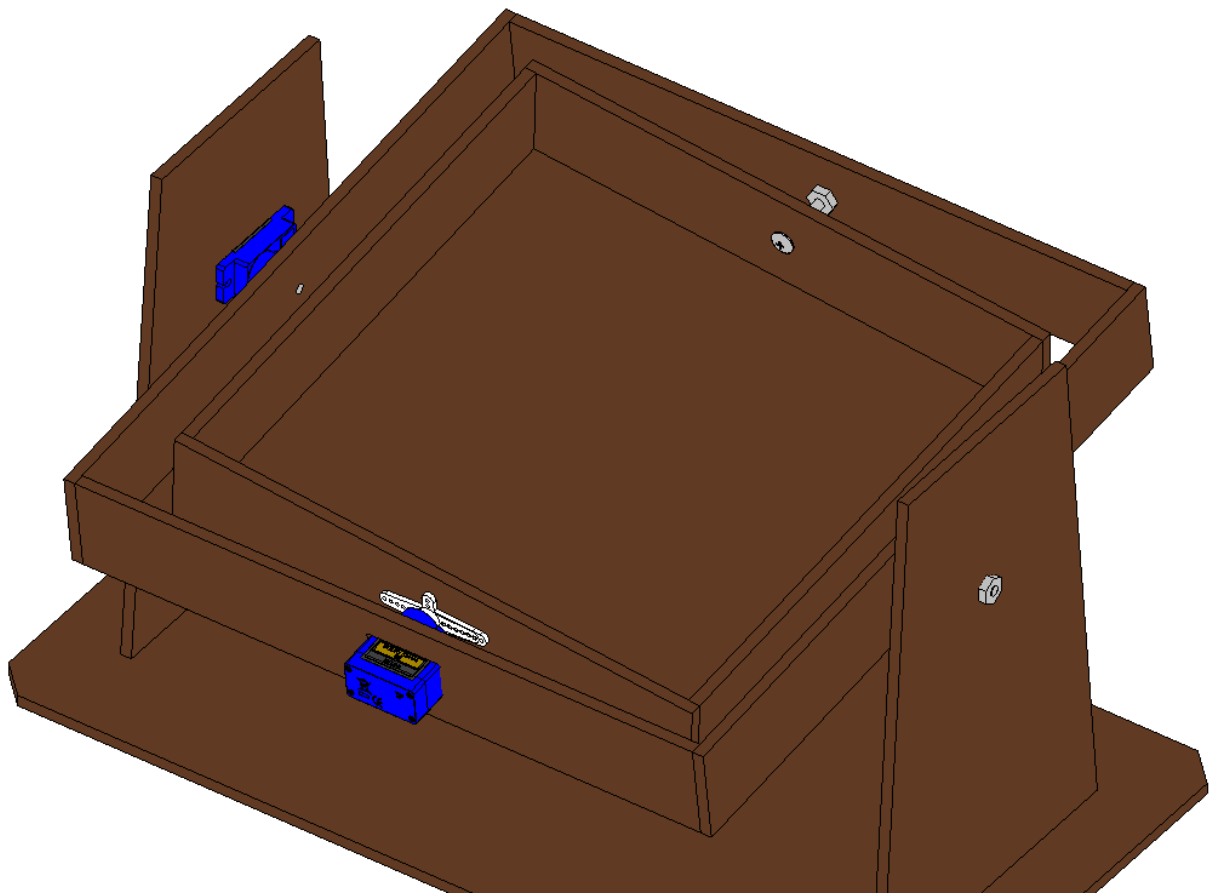
To join these parts together, we will use a **9g Servo** and a **m3 machine screw** (as an axle). To see precisely where each piece joins together, watch the **exploded assembly animation**.

A few points to consider:

- The Servos themselves shouldn't need to be glued, but just pressure fit into their cutouts.
- The Servo horns, however, should be glued or taped into place
 - There will be a centerpoint hole on each piece you attach a servo horn to, use it to align the center of the servo horn.
 - If when you plug this into the servo the fit is too loose, you may use a small screw (check with your instructor to make sure you have the correct one) to fix the horn to the servo output shaft.
 - Don't overtighten this connection – doing so will inhibit the servo movement and potentially damage it
- When using the **m3 machine screw** to create each of the two axles, you'll need to also use three m3 nuts to ensure the spacing between the Inner Maze and Outer Frame is maintained. You can see the preferred arrangement below:



- Each axle needs to be opposite to one of the two Servos.



Congratulations, the build is complete! Now move on to wiring and coding.